

Shivashant Manohar

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EDUCATION

VIT-AP University

2023 - 2027

B.Tech in Computer Science — Artificial Intelligence & Machine Learning CGPA: 8.05/10 Amaravati, Andhra Pradesh

TECHNICAL SKILLS

Languages: Python, SQL, C++

Data Analysis: pandas, NumPy, SQLite, relational schema design, backtesting, matched-control experiments, Matplotlib, Plotly

ML / Deep Learning: PyTorch, scikit-learn, Hugging Face (Transformers, tokenizers, datasets)

Engineering: FastAPI, REST APIs, Streamlit, Gradio, Git, Linux, cron, GitHub Actions, Google Cloud Platform

Audio / Retrieval: librosa, SciPy, music information retrieval, dynamic time warping, FAISS

PROJECTS

Coil — NSE Stock Screener and 10-Year Backtest | *Python, pandas, SQLite, FastAPI*

2025 - 2026

- Showed that a popular chart pattern has no demonstrable edge (11.0% vs 22.3% CAGR for matched random entries, 97 vs 94 trades, costs deducted) by building a ten-year point-in-time backtest that replays 2,657 signals against same-stock, same-risk random controls on a survivorship-controlled top-500 universe.
- Narrowed ~2,300 NSE equities to a handful of daily candidates in ~90s cold and ~6s cached by building a tiered pipeline that loads price, fundamentals and shareholding data into SQLite and applies YAML-defined filters, calling the network only for technical survivors.
- Shipped a live web app (coil-vcp.vercel.app) backed by 400+ tests over ~15k lines of Python by serving a FastAPI backend on Render, a vanilla JavaScript frontend on Vercel, and a nightly GitHub Actions job that refreshes prices and publishes a data snapshot.
- Removed silent drift between the backtest and the live app by consolidating three duplicate trade simulators into one side-effect-free module shared by the study, the portfolio simulator and the app.

Cover Song Retrieval — Music Information Retrieval (*in progress*) | *Python, NumPy, pytest*

2026 - Present

- Built the classical cover-song retrieval baseline for the Da-TACOS benchmark (13,000 tracks, 1,000 cover cliques), validated by 20 unit tests against hand-computed cases, by implementing Optimal Transposition Index key alignment, cosine-cost subsequence DTW and candidate ranking.
- Wrote MAP, MRR and Recall@K from scratch and tested average precision against rankings worked out by hand, so the planned TCN and triplet-loss encoder will be measured against a verified baseline on work-disjoint splits.

Modern Mini LLM — Decoder-Only Transformer from Scratch | *PyTorch, Gradio*

2026

- Trained a ~3.3M-parameter decoder-only language model on TinyStories by implementing RMSNorm, rotary position embeddings, grouped-query attention (8 query / 2 KV heads) and SwiGLU in PyTorch, with a 2,000-token byte-level BPE tokenizer and a memory-mapped token cache.
- Made the model usable end to end with CLI generation (temperature, seed and length controls) and an interactive Gradio demo.

Neural Network from Scratch | *NumPy*

2026

- Reached 95.3% training accuracy on a 3-class spiral dataset that random weight search could not fit by implementing dense layers, ReLU, softmax, categorical cross-entropy and backpropagation in NumPy, then SGD with learning-rate decay and momentum.

Audio Explorer — Interactive Audio AI Preprocessing | *librosa, Streamlit, Plotly*

2026

- Built a Streamlit app that turns WAV, MP3 and FLAC uploads into six inspectable views (waveform, signal health, STFT and 128-band Mel spectrograms, 13 MFCCs with deltas, beat tracking) using librosa, NumPy, SciPy and Plotly.
- Kept feature computation separate from plotting so every feature is reusable outside the UI, and published a first-principles write-up of the pipeline on my portfolio.

RELEVANT COURSEWORK

Data Structures & Algorithms · Database Management Systems · Machine Learning · Deep Learning
Artificial Intelligence · Probability & Statistics · Computer Networks

LEADERSHIP & COMMUNITY

Google Developer Groups VIT-AP — Cloud & DevOps Team

Aug 2024 - Present

Worked with GCP and cloud-native tooling; contributed to developer workshops and community sessions.